

Assignment 3

Positional Encoding & Sequence Order

Strict Protocol Edition · Towards Research-Level Competence

Assignment 3

■ GLOBAL RULES

- Form a hypothesis BEFORE every experiment.
- Explain your reasoning before asking questions.
- After using external help, re-implement from memory.
- Every experiment must include: Hypothesis → Result → Interpretation.

Problem Statement

Prove empirically whether your attention model is position-sensitive or position-invariant, then fix it. The task is sorting: given a sequence of 8 integers, output them in ascending order. You must implement three PE variants and compare them rigorously.

- Input: unsorted sequences of 8 integers from vocab size 32
- Task: output the sorted sequence in ascending order
- Three runs: (1) no positional encoding, (2) sinusoidal PE, (3) learned PE
- Embedding dim ≤ 32 , single attention layer, no residuals or LayerNorm
- Measure: does the model learn order, or does it succeed by ignoring position?

Implementation Requirements

1. Implement sinusoidal PE from scratch using the original Transformer formula
2. Implement learned PE as a trainable embedding lookup table
3. Implement the no-PE baseline -- identical architecture, no positional signal
4. Run all three on the same dataset split and log metrics identically
5. Probe the no-PE model: shuffle input order at test time and measure accuracy drop
6. Visualize the learned PE vectors and compare their geometry to sinusoidal PE

Instrumentation Requirements

1. Log output accuracy vs. position index -- do early/late positions learn faster?
2. Plot attention matrices for all three models at convergence
3. Track gradient flow into the PE parameters (learned) vs. input embeddings (sinusoidal)
4. Measure how fast each variant reaches 90% accuracy -- sample efficiency comparison

Failure Analysis (Required)

You MUST document failures. Describe what failed, why you think it failed, test at least two fixes, and evaluate which fix worked and why.

1. Does the no-PE model succeed? If yes, explain the mechanism it uses.
2. Does sinusoidal or learned PE converge faster? Why do you think that is?
3. What happens if you initialize learned PE to sinusoidal values -- does it help?
4. Test: add noise to PE at test time. How sensitive is each model?

Mathematical Integration

1. Derive the sinusoidal PE formula and explain what frequency bands encode
2. Analyze the dot products between PE vectors at different positions -- what pattern emerges?
3. Compute the rank of the learned PE embedding matrix at convergence
4. Relate PE geometry to the attention score distribution -- can you predict which positions attend to which?

Research Questions to Answer

- Is position information necessary for sorting, or can content alone solve it?
- What does the attention matrix look like in the no-PE model vs. PE models?
- Do PE vectors cluster by position or by value after training?
- What is the minimum sequence length where positional encoding becomes critical?

Phase Progression

Phase	Focus
Phase 1	Build the core architecture
Phase 2	Compare with baseline / non-attention models
Phase 3	Study optimization dynamics
Phase 4	Modify loss functions and evaluate

Phase 5	Alter architecture systematically
Phase 6	Design novel training techniques

Document everything like a research paper. Your goal: independently analyze neural architectures, diagnose training failures, and propose novel improvements grounded in theory and empirical evidence.